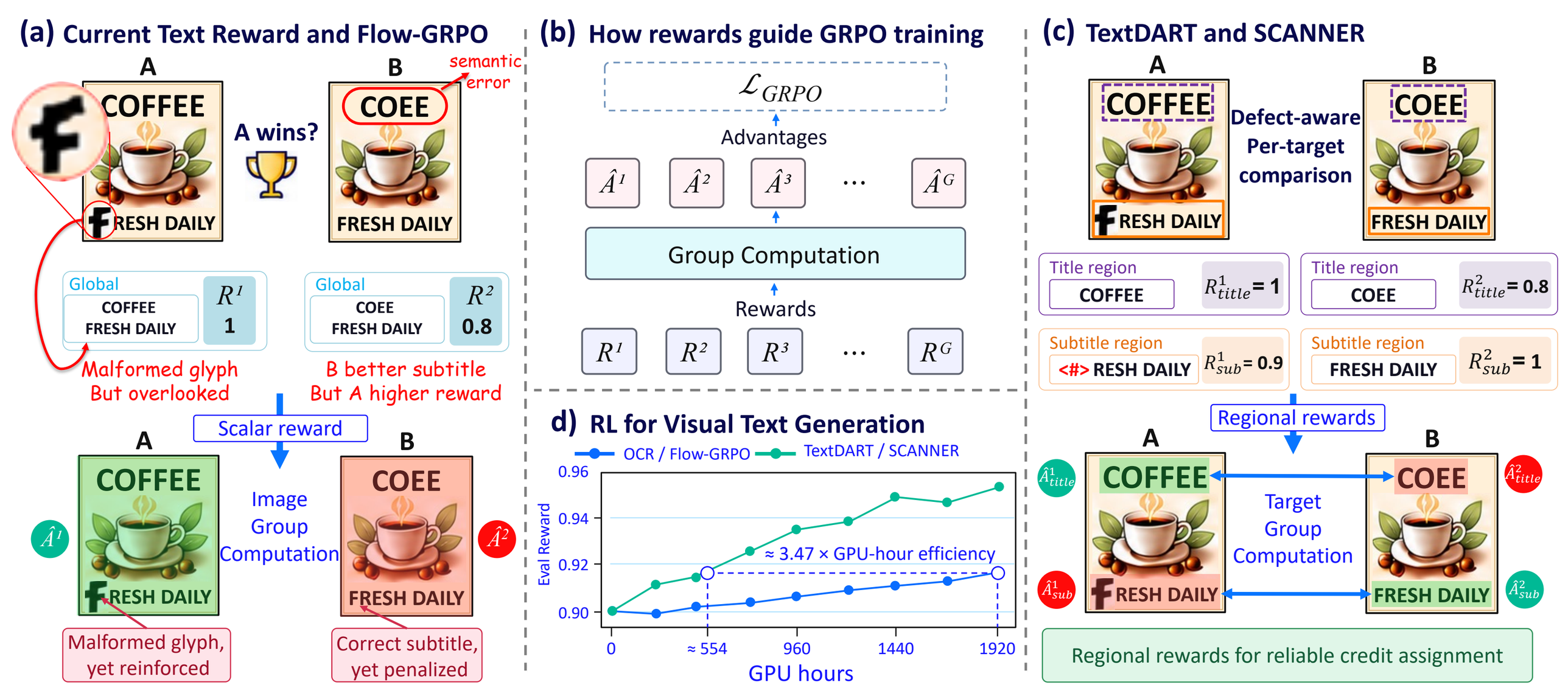

(a) Current Text Reward and Flow-GRPO
A
B
COFFEE
COEE
FRESH DAILY
semantic error
A wins?
Global
COFFEE
FRESH DAILY
R^1
1
Global
COEE
FRESH DAILY
R^2
0.8
Malformed glyph
But overlooked
B better subtitle
But A higher reward
Scalar reward
Image
Group
Computation
$\hat{A}^1$
$\hat{A}^2$
Malformed glyph,
yet reinforced
Correct subtitle,
yet penalized
(b) How rewards guide GRPO training
$\mathcal{L}_{GRPO}$
Advantages
$\hat{A}^1$ $\hat{A}^2$ $\hat{A}^3$ … $\hat{A}^G$
Group Computation
Rewards
R^1 R^2 R^3 … R^G
d) RL for Visual Text Generation
OCR / Flow-GRPO
TextDART / SCANNER
Eval Reward
0.96
0.94
0.92
0.90
≈ 3.47 × GPU-hour efficiency
0
≈ 554
960
1440
1920
GPU hours
(c) TextDART and SCANNER
Defect-aware
Per-target
comparison
Title region
COFFEE
R^1_{title}= 1
Title region
COEE
R^2_{title}= 0.8
Subtitle region
<#> RESH DAILY
R^1_{sub}= 0.9
Subtitle region
FRESH DAILY
R^2_{sub}= 1
Regional rewards
$\hat{A}^1_{title}$
$\hat{A}^1_{sub}$
Target
Group
Computation
$\hat{A}^2_{title}$
$\hat{A}^2_{sub}$
Regional rewards for reliable credit assignment